Lecture 6: Statistical inference – Estimation. Part 2

Yevgen Ryeznik

2019-11-29

Bayesian approach

Let θ be a parameter of interest. Somehow, a prior information about θ is available (historical reports, similar experiments, subjective estimates, etc.). We can represent this information in terms of a prior distribution

$$p(\theta)$$

Then, after performing an experiment where we observe a random variable $Y = y$. Bayes' formula tells that

$$p(\theta|y) = \frac{p(\theta, y)}{p(y)} = \frac{p(\theta)p(y|\theta)}{p(y)}$$

where $p(\theta, y)$ is a ***posterior distribution*** and

$$p(y) = \sum_{\theta} p(\theta)p(y|\theta) \quad \text{(discrete case)}$$

or

$$p(y) = \int_{\Theta} p(\theta)p(y|\theta)d\theta \quad \text{(continuous case)}$$

Formulas can be rewritten as

$$p(\theta|y) \propto p(\theta)p(y|\theta).$$

Example 1 (Spelling correction)

What about the word "*radom*"? It could be:

- Misspelling or mistyping of "random".
- Misspelling or mistyping of "radon".
- Intentional typing of "radom".

Label y as the data (observation of Y) and θ as the word that a person was intending to type:

$$\Pr(\theta|Y = \textit{"radom"}) \propto p(\theta)\Pr(Y = \textit{"radom"}|\theta).$$

Consider only three possibilities for the intended word, θ: ("random", "radon", "radom").

Posterior probability of interest, e.g. for θ_1="random":

$$\Pr(\theta_1|\,''radom'') = \frac{p(\theta_1)\Pr(\,''radom''|\theta_1)}{\sum_{j=1}^{3} p(\theta_j)\Pr(\,''radom''|\theta_j)}.$$

We need:

- Prior probabilities $p(\theta_j)$ (frequencies of words in some large database)
- Likelihoods $p(y|\theta_j)$ (from model of spelling and typing errors)

Remark. *Radom* is a city in Poland

Estimation of probability $p = \mathbf{Pr}(B)$ of some event B

We want to study $p = p_B$. Two approaches can be applied:

- ***Frequentist***. $\widehat{p}_B = \frac{k}{n}$, where k is a number of times when B is true in n trials. The uncertainty on estimate can be done with confidence intervals.
- ***Bayesian***. Uncertainty of p_B is modeled by a p.d.f. $f(p)$. Then

$$\Pr^{\text{pred}}(B) = \mathbf{E}[p] = \int pf(p)dp.$$

The crucial point in Bayesian methodology is the choice of the density $f(\cdot)$. The density reflects all our knowledge about the studied problem.

Combining data and prior knowledge.

- An experiment starts with a ***prior*** assumption about the parameter of interest:

$$f^{\text{prior}}(\theta)$$

- Then, a result of experiment is observed $\Leftrightarrow$ some event C is true. - Prior information is updated by the Bayes' formula:

$$f^{\text{post}}(\theta) = cP(C|\theta)f^{\text{prior}}(\theta)$$

where

$$\frac{1}{c} = P(C) = \int_{-\infty}^{+\infty} P(C|\theta)f^{\text{prior}}(\theta)d\theta.$$

Example 2 (severe adverse event (AE) in a clinical trial)

Let us consider a clinical trial where 1 severe AE out of 3 AEs has been observed, and we are interested in probability p_B of a severe AE. Since we had no information about it before the experiment, we may assume that $p_B \sim U(0,1)$.

$$f^{\text{prior}}(p) = 1.$$

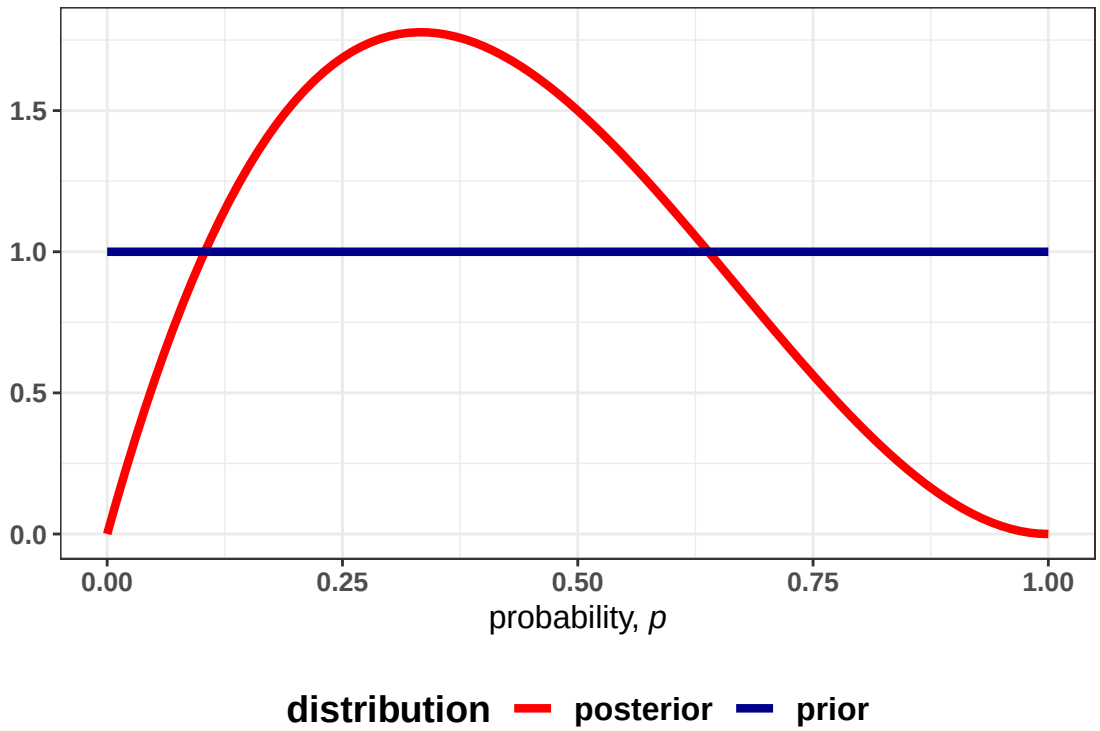
Now, we have observed a severe AE:

$$C = \text{"one severe AE out of 3 AEs"} \Leftrightarrow N = 1,\ N \sim Bin(3, p_B)$$

$$P(N = 1|p) = P(C|p) = \binom{3}{1} p(1-p)^2 \Rightarrow$$
$$f^{\text{post}}(p) = c \cdot 3p(1-p)^2 \cdot 1 = cp(1-p)^2,\ 0 \le p \le 1,$$

which is $Beta(2, 3)$:

$$\theta \sim Beta(a, b), \quad f(\theta) = c\theta^{a-1}(1-\theta)^{b-1}, \quad a, b > 0, \theta \in (0, 1).$$



Credibility Intervals

$(1-\alpha) \cdot 100\%$ credibility interval for the parameter θ is given by

$$\Pr(\theta_{1-\alpha/2} \le \theta \le \theta_{\alpha/2}) = 1 - \alpha.$$

Example 3 (Normal Distribution)
$X_1, X_2, \ldots, X_n \sim N(\mu, \sigma^2)$, and we are interested in estimating μ:

- prior distribution $\mu \sim N(\mu_0, \sigma_0^2)$

It can be shown that the posterior for μ is also normal $N(\mu_1, \sigma_1^2)$ where

$$\mu_1 = \frac{\frac{n}{\sigma^2}\overline{X} + \frac{1}{\sigma^2}\mu_0}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}} \qquad \sigma_1^2 = \left(\frac{n}{\sigma^2}^2 + \frac{1}{\sigma_0^2}\right)^{-1}.$$

95% credibility interval is

$$\mu_1 - 1.96\sigma_1 \leq \mu \leq \mu_1 + 1.96\sigma_1.$$

Choice of a Model for Data: Conditional Distribution

Important step in Bayesian modeling: assume that $F(x;\theta)$ is actually the conditional distribution of x given θ, i.e.

$$\Pr(X \leq x|\theta) = F(x;\theta)$$

The p.d.f. or p.m.f. of the conditional distribution $F(x|\theta)$ is denoted by $f(x|\theta)$.

Example 4 (Tossing a coin)

Fair coin: $X \sim Bin(n, 1/2)$.

Unknown coin: Distribution $P(X \leq x|\theta)$; the p.m.f. is $p_{X|\theta}(x|\theta) = Bin(n, \theta)$.

Conditional Independence

The independence between variables is assumed to be valid for any value of θ

$$\Pr(X_1 \leq x, \ldots, X_n \leq x) = F(x_1|\theta) \cdot \ldots \cdot F(x_n|\theta)$$

for all θ.

Bayesian Updating and Likelihood Functions

Observations: $x_1, \ldots, x_n$ of $X_1, \ldots, X_n$.

Assume that $f(x|\theta)$ is known and that observations of X_i are conditionally independent.

Then,

$$f(x_1, \ldots, x_n|\theta) = f(x_1|\theta) \cdot \ldots \cdot f(x_n|\theta) = \mathcal{L}(\theta)$$

is a likelihood.

$$f^{\text{post}}(\theta) = c\mathcal{L}(\theta)f^{\text{prior}}(\theta), \qquad \frac{1}{c} = \int_{-\infty}^{+\infty} \mathcal{L}(\theta)f^{\text{prior}}(\theta)d\theta.$$

Conjugate Priors

A conjugate prior is an algebraic convenience, giving a closed-form expression for the posterior; otherwise a difficult numerical integration may be necessary.

Further, conjugate priors may give intuition, by more transparently showing how a likelihood function updates a distribution.

Beta distribution

$$\theta \sim Beta(a,b), a,b>0, \text{ and } f(\theta) = c\theta^{a-1}(1-\theta)^{b-1},\ 0 \le \theta \le 1, c = \tfrac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}$$
$$E[\theta] = p, V[\theta] = \tfrac{p(1-p)}{(a+b+1)}, \text{ where } p = \tfrac{a}{a+b}.$$

The Beta priors are conjugated priors for the problem of estimating the probability $p = \Pr(B)$.

- Observed in n tries: the statement B was true k times.
- Prior density: $f^{\text{prior}}(\theta) \sim Beta(a,b)$
- Posterior density follows as with

$$f^{\text{post}} = Beta(\widetilde{a}, \widetilde{b}), \text{ where } \widetilde{a} = a+k, \widetilde{b} = b+n-k.$$

The $Beta(a,b)$ prior is equivalent to observing that the event B has occurred a times in $a+b$ experiments.

Dirichlet distribution

$$\theta = (\theta_1, \theta_2) \sim Dirichlet(a_1, a_2, a_3), a_i > 0$$
$$f(\theta_1, \theta_2) = c\theta_1^{a_1-1}\theta_2^{a_2-1}(1-\theta_1-\theta_2)^(a_3-1), \theta_i > 0, \theta_1+\theta_2 < 1$$
$$c = \tfrac{\Gamma(a_1+a_2+a_3)}{\Gamma(a_1)\Gamma(a_2)\Gamma(a_3)}$$
$$\mathbf{E}[\theta_i] = \tfrac{a_i}{a_0}, \mathbf{V}[\theta_i] = \tfrac{a_i(a_0-a_i)}{a_0^2(a_0+1)}, \text{ where } a_0 = a_1+a_2+a_3, i = 1,2.$$

The marginal distributions: $\theta_i \sim Beta(a_i, a_0 - a_i)$, $i = 1,2$.

The Dirichlet priors are conjugated priors for the problem of estimating the probabilities $p_i = \Pr(B_i)$, $i = 1,2,3$, such that B_i are disjoint and $p_1 + p_2 + p_3 = 1$.

Let $\theta_i = p_i$.

- Observed in n tries: the statement B_i was true k_i times.
- The prior density $f^{\text{prior}}(\theta_1, \theta_2) \sim Dir(a_1, a_2, a_3)$.
- Posterior density:

$$f^{\text{post}}(\theta_1, \theta_2) \sim Dir(\widetilde{a}_1, \widetilde{a}_2, \widetilde{a}_3), \text{ where } \widetilde{a}_1 = a_1+k_1, \widetilde{a}_2 = a_2+k_2, \widetilde{a}_3 = a_3+k_3.$$

Gamma distribution

$$\begin{gathered}\theta \sim \Gamma(a,b), a,b>0,\\ f(\theta) = c\theta^{a-1}(1-\theta)^{b-1}, 0 \le \theta \le 1,\\ c = \tfrac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\\ \mathbf{E}[\theta] = \tfrac{a}{b}, \mathbf{V}[\theta] = \tfrac{a}{b^2}.\end{gathered}$$

The Gamma priors are conjugated priors for the problem of estimating the Poisson λ.

- Observed: Observed in n tries (Poisson experiment): the statement B was true k times.
- Prior density: $f^{\text{prior}}(\theta) \sim \Gamma(a,b)$.
- Posterior density:

$$f^{\text{post}}(\theta) \sim \Gamma(\widetilde{a},\widetilde{b}), \text{ where } \widetilde{a} = a+k, \widetilde{b} = b+n.$$